From Biochemical Hits to Drug-Like Leads: A Critical Evaluation of Heparanase Inhibitor Discovery

Abstract

Heparanase (HPSE1) remodels heparan sulfate in extracellular and pericellular environments and contributes to processes associated with malignancy, inflammation, and vascular injury. Numerous inhibitors have demonstrated biochemical activity, yet progression toward clinically useful drugs has been limited. This critical narrative review examines whether evidence for representative inhibitor classes supports progression from enzyme activity to pharmacologically credible leads. Drawing on primary experimental studies, structural research, and relevant reviews identified through focused searches, it considers early heterocyclic inhibitors, benzazole and triazolo-thiadiazole series, tetrahydroimidazopyridine optimization, mechanism-based cyclophellitols, fragment discovery, defined glycomimetics, polymeric inhibitors, and clinical investigations of heparan sulfate mimetics. Three conclusions emerge. First, enzyme potency depends strongly on substrate and assay architecture, making unqualified cross-study IC_{50} rankings unreliable. Second, selectivity, direct binding evidence, cellular target engagement, and pharmacokinetic measurements remain unevenly characterized across chemotypes. Third, preclinical efficacy does not by itself establish an HPSE-specific mechanism, and early clinical tolerability has not consistently translated into meaningful benefit. Recent crystallographic fragments, modified glycans, and covalent inhibitors provide new strategies but remain at different stages of development. Cross-chemotype comparison shows that matched structural changes can have context-dependent, even reversed, effects and that inhibitor classes offer complementary evidence profiles rather than a universal potency hierarchy. The review connects these medicinal chemistry findings to assay context, target engagement, exposure, and clinical outcomes. Future programs should prioritize orthogonal assays, modality-appropriate selectivity, and exposure-linked pharmacodynamics.

Keywords: Heparanase; HPSE1; heparan sulfate; inhibitor discovery; medicinal chemistry; structure–activity relationship; selectivity; target engagement; pharmacokinetics; translational pharmacology.

1 Introduction

Heparan sulfate proteoglycans connect extracellular matrix organization to growth-factor signaling, inflammatory-cell movement, tissue repair, and vascular barrier function. Their heparan sulfate chains organize interactions with proteins rather than acting only as passive structural components. Heparanase-1 (HPSE1), classically described as a mammalian endo-β-D-glucuronidase, cleaves these chains and can alter both the physical integrity of tissue barriers and the availability of heparan sulfate-associated signaling molecules. Structural studies established a catalytic machinery centered on Glu225 and Glu343, whereas recent experiments with defined substrates suggest that cleavage behavior can depend on substrate sequence and context [1, 26, 37].

Abnormal heparanase expression or activity has been associated with tumor invasion, angiogenesis, metastatic dissemination, treatment resistance, inflammation, and microvascular dysfunction. These associations have motivated inhibitor programs using sulfated polysaccharides, oligosaccharides, synthetic glycomimetics, heterocyclic small molecules, irreversible glycosidase-directed inhibitors, and more recently induced protein degradation. The mechanistic breadth of heparanase biology creates opportunity but complicates interpretation. Reduced cellular invasion after compound exposure could reflect direct catalytic inhibition, disruption of growth-factor binding, altered receptor signaling, nonspecific charge-dependent interactions, or combinations of these mechanisms [3, 5, 44].

The historical emphasis on enzyme potency has produced a diverse chemical literature, while the clinically tested agents have largely been heparan sulfate mimetics. Zhang and Cui [2] reviewed small-molecule discovery and screening, but the field still lacks an integrated comparison connecting within-series medicinal chemistry to assay context, experimental mechanism, modality-matched selectivity, cellular and animal pharmacology, exposure, and clinical outcomes. These dimensions are often reported in different studies and are rarely assessed together to explain why one series advances while another remains a biochemical hit. This review adds that cross-stage, cross-chemotype interpretation, without assigning a universal score or treating molecular size as a proxy for drug-likeness. Small molecules remain the primary focus; chemically defined and clinically investigated glycomimetics serve as comparators whose specific selectivity and pharmacological properties are evaluated rather than presumed.

This critical narrative review integrates medicinal chemistry, biochemical, biophysical, pharmacological, and clinical literature. Focused searches of PubMed, Scopus, ScienceDirect, and Google Scholar combined terms for heparanase or HPSE with inhibitor discovery, heparan sulfate mimetics, structure–activity relationships, fragment screening, and structure-based design. The discussion emphasizes literature published from 2020 through 2026, alongside earlier studies that established the field's structural, chemical, and clinical foundations. Studies were selected for their relevance to inhibitor chemistry and evidence on biochemical activity, binding mechanism, selectivity, cellular or animal activity, pharmacokinetics, or clinical outcomes. Findings are synthesized thematically, with potency interpreted in the context of each study's assay. The literature search was last updated on 25 October 2026. As a focused narrative synthesis, selection was qualitative rather than systematic; no screening counts or pooled comparative estimates are claimed. The synthesis follows evidence from biochemical inhibition and its confirmation through selectivity, mechanism, cellular relevance, and pharmacological translation; these complementary lines organize comparison across chemotypes but are not combined into a numerical lead score.

2 Structural and biochemical context for inhibitor design

2.1 Catalytic site and substrate recognition

Active human HPSE1 is generated through proteolytic maturation of a precursor and contains associated subunits forming the catalytic architecture. Glu225 acts as the catalytic acid/base, and Glu343 participates as the nucleophile in glycosidic bond cleavage. The groove surrounding the active site accommodates sulfated carbohydrate motifs through a network of electrostatic, polar, and hydrophobic contacts. The crystal structures analyzed by Zhang et al. [26] include enzyme complexes with a heparin-derived tetrasaccharide and with small molecules, providing a basis for discussing how a candidate ligand may recognize either catalytic or adjacent substrate-binding regions.

Substrate recognition is not governed by negative charge alone. Sulfation position, uronic-acid configuration, chain length, conformational presentation, and accessibility can affect substrate turnover and ligand recognition. Wang et al. [37], using structurally defined oligosaccharides, reported substrate-dependent cleavage behavior that complicates an exclusively endo-acting description of the enzyme. This result is particularly relevant to inhibitor studies: assays using short synthetic substrates, fondaparinux-related fragments, fluorescent probes, or matrix-embedded heparan sulfate are not necessarily reporting inhibition under equivalent molecular conditions.

2.2 Beyond a single catalytic pocket

A compound detected in the catalytic cleft may inhibit by blocking substrate access or engaging residues required for catalysis, but crystallographic binding alone does not establish functional inhibition. Davies et al. [25] used fragment screening and structural analysis to identify 31 binding fragments, of which five inhibited enzymatic activity in the micromolar range. Their observations included orthosteric and allosteric binding opportunities. A fragment grown through two iterative steps exhibited approximately sevenfold improved potency relative to its starting fragment. The importance of this study lies not in demonstration of an optimized therapeutic lead, but in experimentally showing that sites outside the traditional substrate-mimetic design space can support ligand binding and chemical optimization.

The mechanistic work of Whitefield et al. [28] on pentosan polysulfate adds a different lesson. Inhibition was associated with multiple overlapping binding events affected by oligomer length and protein structural changes. A simple one-site competitive model is therefore not a universal description of heparanase inhibitor action. Mechanism assignment should be linked to inhibitor class, assay design, and kinetic evidence rather than inferred from a single docking pose.

3 Inhibitor chemotypes and evidence quality

3.1 Heparan sulfate mimetics as development benchmarks

3.1.1 Clinically investigated sulfated agents

Molecularly complex and highly sulfated agents provided some of the earliest advanced pharmacological tests of the HPSE hypothesis. Muparfostat (PI-88), roneparstat (SST0001), pixatimod (PG545), and necuparanib (M-402) entered clinical development in different disease settings. Their mechanisms are not identical, and several affect heparan sulfate-binding growth factors or other pathways in addition to HPSE activity. Describing the entire group as uniformly heterogeneous or promiscuous obscures distinctions in chemical definition, sulfation pattern, pharmacokinetics, and interaction profiles.

Ferro et al. [4] showed that incorporating lipophilic aglycone functionality into sulfated oligosaccharide glycosides could improve pharmacological features, including preliminary intravenous half-life, while maintaining inhibition of heparanase and angiogenic growth-factor signaling. The resulting PG545 series illustrates how deliberate molecular design can improve exposure properties even within a sulfated glycan

scaffold. Dredge et al. [5] subsequently documented antitumor and antimetastatic activity in preclinical models; Ritchie et al. [3] showed that SST0001 interfered with heparanase/syndecan-1 biology and reduced myeloma growth and angiogenesis. Such results supply genuine disease-model pharmacology, although polypharmacology complicates assignment of the total antitumor benefit to HPSE catalysis alone.

3.1.2 Defined glycans, sulfation pattern, and multivalency

The progression from complex polysaccharide mixtures to more structurally defined glycomimetics should not be framed simply as replacing glycans with small molecules. Zubkova et al. [30] reported discrete dendrimer glycomimetics with low-nanomolar enzyme inhibition, reduced tumor burden in a myeloma xenograft model, and favorable reported cytotoxicity and anticoagulant observations within the tested conditions. Loka et al. [31] systematically varied the sulfation pattern of defined glycopolymers, combined enzymatic testing with binding assays against other heparan sulfate-binding proteins, and demonstrated antimetastatic activity in a mouse model. These findings show that chain presentation and sulfation topology influence biochemical and biological selectivity.

Shi et al. [32] investigated sulfated hyaluronan, reporting HPSE binding, reduced enzymatic activity, inhibition of carcinoma-cell invasion, and modulation of chemokine-associated signaling. Wakpal et al. [33] used aminoglycosides as accessible starting materials for a library of defined heparan sulfate-like oligosaccharides; the work particularly addressed HPSE interaction relative to platelet factor 4. Hotor et al. [34] extended aminoglycoside-derived structures and reported greater than 300-fold selectivity in the specified comparative protein-binding assays for selected ligands, while Abdulsalam et al. [35] linked sulfated and hydrophobic glycan chemistry to tumor-model activity. Ni et al. [49] found that synthetic glycol-split heparin trisaccharides were more potent than intact uronic-acid analogues, with STD-NMR and docking results aligning with enzyme-assay findings. Sugimoto et al. [50] reported a synthetic fucoidan analogue with approximately threefold greater HPSE inhibitory activity than its comparator and no measured factor Xa inhibition; it also bound SARS-CoV-2 spike and inhibited viral infection in cell assays. Because the antiviral mechanism can involve spike binding as well as HPSE inhibition, those results do not by themselves establish HPSE-dependent cellular target engagement. These are substantive design advances; nevertheless, limited off-target panels and selected cell-line results should not be generalized to comprehensive safety or drug disposition.

Wang et al. [45] subsequently applied 3D-QSAR to this aminoglycoside-derived series and profiled lead L17 in HPSE-dependent cancer cell lines. They reported concentration-dependent effects on proliferation and invasion, reduced extracellular HPSE levels, and minimal CYP3A4 inhibition in an experimental assay despite an in-silico liability flag. This extends the evidence from enzyme inhibition toward cellular activity and a limited metabolic counter-screen, but does not establish in-vivo exposure, systemic safety, or efficacy.

Recent polymeric mimetics provide an additional check against the presumption that molecular miniaturization is always the preferred path. Singh et al. [36] reported myeloma-model activity for an HPSE-inhibiting polymeric mimetic. Teki et al. [38] examined sulfated, sulfonated, and multivalent thioglycoside designs. Wang et al. [37] synthesized defined heparan sulfate oligosaccharides, probed substrate specificity, and reported a submicromolar inhibitor. Together, these investigations indicate that charge placement, structural definition, multivalency, and substrate mimicry should be assessed as independent design variables. Their apparent biochemical advantages remain distinct from proven systemic bioavailability or clinical benefit.

3.2 Small-molecule chemotypes and experimentally supported SAR

3.2.1 Early heterocyclic inhibitor series

The synthetic small-molecule field predates modern HPSE structural biology. Courtney et al. [11, 12] disclosed isoindole and heteroaryl acetic-acid derivatives, establishing medicinal chemistry starting points without requiring large sulfated carbohydrate scaffolds. Rondanin et al. [14] investigated arylamidonaphthalene sulfonates inspired partly by antimetastatic suramin-related chemistry and described inhibitory activity alongside selected cellular effects. These reports established chemical feasibility but did not provide the uniform selectivity, exposure, and target-engagement datasets now expected in a robust lead-optimization campaign.

Messore et al. [15] provide unusually interpretable within-series SAR for symmetrical benzazolyl inhibitors measured with a fondaparinux-based HPSE assay. In the nonfluorinated pair, replacing the urea linker in 5a (IC_{50} = 3.20 μM) with thiourea in 5d improved potency to 1.81 μM. The same linker change had the opposite effect in the fluorinated pair: urea-containing 5c inhibited at 0.18 μM, whereas thiourea 5e inhibited at 1.40 μM. Fluorination itself improved the urea analogue from 5a to 5c by about 18-fold. Extending the terminal carboxymethyl groups of 5e to glycine amides in 7g was associated with a further 17.5-fold potency increase (1.40 to 0.08 μM). Thus, the effect of linker identity depends on the surrounding substitution pattern, while the glycine extension is a promising within-assay matched-pair result rather than a general rule about polarity or charge. Figure 2 displays the corresponding 2D structure depictions and same-assay activity values. Compound 7g was less potent in that assay than the roneparstat comparator (0.005 μM). Selected compounds also reduced invasion and altered proangiogenic gene-expression readouts, but those phenotypes do not by themselves establish cellular HPSE target engagement or clinical exposure.

Pala et al. [16] expanded scaffold diversity through iterative ligand-based virtual screening, followed by experimental testing of 55 compounds, with 19 showing IC_{50} values below 5 μM in the reported assay. The identification of terephthalamide and isophthalamide-type scaffolds highlights the value of systematic chemical-space exploration. It does not, by itself, establish that scaffold novelty, favorable docking, or low-micromolar enzyme inhibition predicts pharmacological tractability.

3.2.2 Triazolo-thiadiazoles and biological efficacy

Baburajeev et al. [17] screened 150 molecules from multiple scaffold families and identified triazolo-thiadiazole compounds with HPSE inhibitory activity. Their testing used radiolabeled extracellular matrix substrate and cell-invasion experiments, strengthening the connection between an enzyme assay and a biologically relevant matrix-based endpoint. Barash et al. [18] further investigated heparanase-inhibiting triazolo-thiadiazoles and reported suppression of primary tumor growth and metastasis in experimental models. These studies are important because they move beyond simple enzyme IC_{50} tables. Nonetheless, tumor growth can be influenced by several concurrent pharmacological actions, and interpretation benefits from direct exposure and HPSE-dependent mechanistic controls.

3.2.3 Tetrahydroimidazopyridines: explicit attention to selectivity

The tetrahydroimidazo[1,2-a]pyridine-5-carboxylic acid program reported by Imai and colleagues is especially pertinent to lead quality. The initial lead 2 showed moderate HPSE1 potency and unresolved selectivity over exo-β-D-glucuronidase and glucocerebrosidase. X-ray-guided and fragment-molecular-orbital-supported optimization produced 4e, with more than sevenfold improved HPSE1 inhibition. Methylation of the 6-hydroxyl group of 4e then yielded compound 18, which reduced inhibition of the comparator enzymes while retaining HPSE1 activity; after dosing, it significantly suppressed mouse serum HPSE1 activity at 3, 30, and 100 mg/kg, with the effect lasting up to 4 h [19–21]. This explicit matched change links potency, enzyme selectivity, and systemic pharmacodynamics, although serum suppression does not establish target engagement in diseased tissue. A companion investigation improved synthetic accessibility, shortening a demanding route by five steps for selected analogues.

This sequence represents a more mature optimization logic than selection on potency alone. Enzyme selectivity, chemical synthesis practicality, direct structure–activity interpretation, and post-dose pharmacodynamic measurement were considered jointly. Nevertheless, the absence of a complete human pharmacokinetic or efficacy dossier means that the compounds should not automatically be described as clinically qualified leads. A favorable serum pharmacodynamic response is a higher level of evidence than docking, but it remains separate from demonstrated target engagement in the pathological tissue and from disease modification.

3.2.4 Mechanism-based cyclophellitols and structurally characterized inhibitors

De Boer et al. [22] appended an HS-like α-1,4-GlcNAc unit to a glucurono-cyclophellitol epoxide, producing pseudodisaccharide inhibitors with nanomolar potency and irreversible HPSE inhibition; they also reported reduced cancer-associated invasion and murine metastasis. Follow-up SAR showed that replacing the GlcNAc motif with glucosamine abolished measurable HPSE inhibition up to 100 μM, while Borlandelli et al. [23] found that selected 4-O-alkyl analogues retained low-micromolar HPSE inhibition and showed improved selectivity over exo-β-D-glucuronidase, supported by structural and computational analysis. These comparisons show that the covalent warhead alone does not determine potency or selectivity. Ehren et al. [24] further expanded functionalized cyclophellitol chemistry. The attraction of covalent mechanism-based inhibition lies in sustained enzyme suppression and potentially high target engagement. Countervailing questions include selectivity among glycosidases, irreversible exposure of off-target proteins, duration of recovery, and suitability of reactive groups for the proposed route of administration.

Chen et al. [27] provided a valuable counterexample to overly simple classifications of reversible and covalent glycosidase ligands. Their crystallographic and kinetic work showed that siastatin B can form distinct inhibitory chemical species, with activity dependent on the enzyme and binding context. The broader lesson is that assigning a ligand's mechanism from its starting chemical structure alone can be misleading; experimental chemistry, kinetics, and structures must be interpreted together.

3.2.5 Fragment binding and emerging chemical space

The 2026 Davies fragment study supplies a contemporary starting point for orthosteric and allosteric designs. It supports actual occupancy of binding sites and experimentally observed micromolar inhibition in a subset

of fragments. The sevenfold optimization improvement is a meaningful within-series result. However, no inference of advanced drug-likeness follows unless additional studies establish physicochemical behavior, selective HPSE engagement, permeability or relevant tissue exposure, and pharmacological responses. In practical terms, a structurally verified fragment is a high-quality discovery starting point but not necessarily a drug-development lead. The representative series in Table 1 illustrate how unevenly these evidence stages have been addressed; potency values remain specific to their source assays.

Table 1: Selected inhibitor series with interpretable experimental evidence. Potency values refer to the source study's assay and should not be compared as if they were obtained under identical conditions. NR indicates that the corresponding evidence is not reported or not established in the examined primary source.

Study and chemotype	**Biochemical evidence**	**Additional experimental evidence**	**Critical interpretation**
Courtney et al. [11, 12], heterocyclic acids	Enzyme inhibition across synthesized analogs	Early SAR	Discovery starting points; downstream pharmacology limited
Messore et al. [15], symmetrical benzazolyl series	5a 3.20; 5d 1.81; 5c 0.18; 5e 1.40; 7g 0.08 μM, fondaparinux assay	Invasion and proangiogenic readouts for selected compounds	Substitution effects are context-dependent; cellular and exposure evidence remains incomplete
Pala et al. [16], virtual-screening hits	19 of 55 tested compounds with IC_{50} below 5 μM	Cellular invasion and gene-expression tests on selected hits	Demonstrates experimental follow-up but not clinical lead qualification
Baburajeev et al. [17], triazolo-thiadiazoles	Direct HPSE activity screening with matrix-associated assay	Invasion experiments	Relevant functional validation; mechanism specificity needs care
Barash et al. [18], triazolo-thiadiazoles	HPSE inhibitory activity	Tumor and metastasis experiments in animals	Preclinical biological relevance, exposure/causality not interchangeable
Imai et al. [19–21], tetrahydroimidazopyridines	Potency improved by structure-guided analog design	Selectivity over GUSβ/GBA; serum HPSE1 pharmacodynamics for optimized compound	Multi-domain lead-optimization evidence
De Boer et al. [22], cyclophellitol derivatives	Nanomolar irreversible HPSE inhibition	Cellular and mouse metastasis outcomes	Strong mechanism-based proof of concept; irreversible selectivity requires evaluation
Borlandelli et al. [23], substituted cyclophellitols	HPSE inhibition and selectivity over exo-β-glucuronidase	Crystallographic validation	Mechanism and selectivity supported by complementary methods
Zhang et al. [26], TC LPA5 4	IC_{50} 10 μM; MX4-62 experimentally inactive	Docking and MD, including computational analog designs	Computational SAR not equivalent to confirmed SAR
Davies et al. [25], fragment series	Five micromolar inhibitory fragments; sevenfold improved potency after growing	Experimental fragment binding structures	Strong discovery foundation, pharmacology at early stage
Zubkova et al. [30], dendrimer mimetics	Low-nanomolar inhibition	Anticoagulant/cytotoxicity tests and myeloma model	Defined glycomimetic with multiple evidence domains

Study and chemotype	Biochemical evidence	Additional experimental evidence	Critical interpretation
Loka et al. [31], defined glycopolymers	Potent enzyme inhibition	Competition against HS-binding proteins; mouse antimetastatic activity	Glycan architecture and selective interactions matter
Hotor et al. [34], tobramycin glycans	Enzyme inhibition in selected analogs	Comparative PF4/antithrombin binding; cell-line activity	Encouraging targeted selectivity, not universal safety
Wang et al. [37], defined HS oligosaccharides	Inhibitory nonasaccharide IC_{50} 0.761 μM	Substrate-cleavage mechanism studies	Important biochemical and mechanistic study; clinical safety not shown
Wang et al. [45], aminoglycoside glycan L17	3D-QSAR-guided follow-up of sulfated and hydrophobic analogs	Cancer-cell proliferation, invasion, extracellular HPSE, CYP3A4 counter-screen	Cellular follow-up is encouraging; in-vivo exposure and efficacy remain unestablished
Ni et al. [49], glycol-split heparin oligosaccharides	Glycol-split trisaccharides more potent than intact analogues	STD-NMR binding data aligned with docking and activity	Useful structure–activity evidence; in-vivo pharmacology not established
Sugimoto et al. [50], synthetic fucoidan analogue	Approximately threefold greater HPSE inhibition than comparator	Factor Xa counter-screen; SARS-CoV-2 spike-binding and cell-infection assays	Antiviral activity may involve spike binding; no in-vivo pharmacology established
Zhou et al. [40], sulfated tetrasaccharides	Compound 5 IC_{50} 11.27 μM	Sepsis survival, inflammatory and morphological outcomes in mice	Moderate assay potency with relevant disease-model effects
Mo et al. [51], low-molecular-weight dextran sulfate	Direct HPSE binding and inhibition; reduced HPSE expression	Invasion, angiogenesis, mouse metastasis and safety assessments	Broad preclinical evidence; exposure and target-specific causality merit follow-up
Crompton et al. [52], OVZ/16-35 glycomimetic	Tetravalent HS mimic	Rat MCD model; glycocalyx protection and reduced albuminuria	PK, bioavailability and long-term safety remain under study; compound patents disclosed

3.3 Why biochemical potency cannot be compared without assay context

3.3.1 Distinct substrates yield distinct measurements

Published assays use natural or radiolabeled extracellular matrix heparan sulfate, heparin-derived fragments, fondaparinux, fluorescence-generating synthetic substrates, and other reporting systems. These methods differ in substrate size, sulfation, matrix presentation, readout physics, turnover conditions, and potential compound interference. A compound may act as a competitive substrate-site blocker under one condition but display apparent mixed inhibition or reduced effectiveness under another. Liu et al. [29] introduced an ultrasensitive fluorogenic probe that greatly improved practical activity detection, whereas Whitefield et al. [28] demonstrated how glycomimetic inhibition can depend on ligand size and multiple binding modes.

The apparent precision of published IC_{50} values must not conceal their experimental context. IC_{50} is conditional on assay parameters and does not equal a universal thermodynamic affinity constant. Direct comparison is most defensible within a common enzyme preparation, substrate concentration, detection system, incubation time, and data-fitting procedure. A reported low-nanomolar value for a multivalent glycopolymer cannot be ranked uncritically against a small molecule measured using a different substrate and platform. For a credible literature table, each potency figure should be accompanied by substrate and assay identifiers whenever available.

Wang et al. [47] developed a single-heparan-sulfate-substrate isotope-dilution mass-spectrometry method (SHS-IDMS) for quantifying HPSE activity in plasma. Using a defined 12-mer substrate and LC-MS/MS product quantification, the method distinguished plasma samples from children with sepsis and healthy children, with activity correlated to syndecan-1 and angiopoietin-2. This is a promising advance for measuring enzyme activity in biological matrices, but it has not yet been established as a standardized inhibitor-response assay across chemotypes or dosing studies. The authors disclosed commercial interests related to glycan therapeutics, making independent replication particularly valuable.

3.3.2 Orthogonal confirmation and interference controls

An initial fluorescence or colorimetric hit should ideally be reproduced using an independent detection method. Counterscreens for optical interference, aggregation, nonspecific polyanion interactions, and effects on the reporter chemistry are especially important for highly charged compounds and chromophoric small molecules. Tests against related glycosidases help assess catalytic-family selectivity, but do not replace checks against unrelated heparan sulfate-binding proteins. Similarly, SPR, NMR, thermal-shift, or crystallographic binding should be interpreted with the understanding that binding does not necessarily inhibit catalysis. For reliable comparison, studies should also report whether inhibition was concentration-dependent, replicated across experiments, supported by orthogonal chemistry, and accompanied by relevant positive and negative controls. When these details are absent or inaccessible, the appropriate interpretation is "not reported in the examined source," rather than an assumption that the study performed or omitted them.

3.4 Selectivity, structural evidence, and target engagement

Selectivity has at least three meanings in HPSE pharmacology. First, a compound may discriminate HPSE1 from other carbohydrate-processing enzymes such as exo-β-D-glucuronidase and glucocerebrosidase. Second, sulfated ligands may interact with platelet factor 4, antithrombin, growth factors, selectins, or numerous heparan sulfate-binding proteins. Third, cellular disease readouts may reflect modulation of both catalytic and noncatalytic HPSE functions. A single selectivity ratio cannot represent all three categories.

Imai et al. [19–21] provide comparatively direct evidence of enzymatic selectivity optimization, whereas Borlandelli et al. [23] combine selectivity measurements with structure-guided interpretation for cyclophellitols. In the glycomimetic literature, Loka et al. [31], Wakpal et al. [33], and Hotor et al. [34] demonstrate that selected binding interactions can be tuned away from particular competing proteins. These studies move the literature beyond the traditional premise that all sulfated ligands necessarily behave alike. At the same time, binding selectivity relative to a few chosen proteins is not equivalent to whole-body pharmacological specificity.

Cellular claims should distinguish a phenotype consistent with HPSE modulation from direct cellular target engagement and from disease-model efficacy. In Messore et al. [15], benzazolyl compounds combined purified-enzyme inhibition with reduced invasion and altered proangiogenic readouts; the cellular findings support biological activity but do not independently establish that HPSE inhibition caused each phenotype. By contrast, the Imai series [19–21] includes a post-dose reduction in mouse serum HPSE activity, a proximal pharmacodynamic observation, although it does not establish engagement in diseased tissue. Loka et al. [31] combined defined sulfation patterns with selected protein-binding comparisons and antimetastatic activity in mice, strengthening a structure-to-function argument without making the animal outcome a direct occupancy measurement. Genetic or rescue controls and structurally distinct inhibitors can sharpen causal attribution when available, but the evidence should be described at the level each experiment actually supports.

4 Translational evidence from models to clinic

4.1 Cellular and preclinical activity: compelling but heterogeneous evidence

Several chemotypes have progressed beyond isolated enzyme assays. The benzazole work of Messore et al. [15], the triazolo-thiadiazole investigations of Baburajeev et al. [17] and Barash et al. [18], and the mechanism-based cyclophellitols of de Boer et al. [22] provide examples of inhibitory chemistry accompanied by cell-based or animal observations. Their results support biological relevance, yet comparisons across tumor models, dosing schedules, invasion assays, and compound exposures remain difficult.

Glycomimetics have generated a particularly broad set of animal observations. PG545 and SST0001 were evaluated in tumor or myeloma models, while defined dendrimers and glycopolymers inhibited metastatic or myeloma-related outcomes in selected mouse experiments. More recent work by Abdulsalam et al. [35] and Singh et al. [36] expands this field through aminoglycoside-derived glycans and polymeric heparan sulfate mimetics. Mo et al. [51] reported that low-molecular-weight dextran sulfate sodium directly bound HPSE, inhibited its activity and expression, reduced 4T1-cell invasion and endothelial angiogenesis, and suppressed pulmonary metastasis in mice. The accompanying in-vivo safety assessment strengthens its preclinical profile, while its polymeric, sulfated scaffold still warrants exposure and target-specificity characterization. These studies should not be viewed as merely historical benchmarks: some provide in-vivo efficacy and off-target-binding investigations that individual newer small-molecule fragments have not yet reached.

Nononcological models extend the translational question. Gamez et al. [41] used heparanase inhibition to investigate endothelial glycocalyx preservation in diabetic microvascular injury, while Wang et al. [39] evaluated CV122 in experimental sepsis. Zhou et al. [40] reported that a structurally simplified sulfated tetrasaccharide, compound 5, inhibited HPSE with IC_{50} = 11.27 μM and improved survival and inflammatory outcomes in a mouse sepsis model. Li et al. [42] and Bai et al. [43] investigated HPSE-targeted approaches in pulmonary fibrosis and chronic obstructive pulmonary disease, respectively. Crompton et al. [52] extended the glycomimetic evidence to a rat model of minimal change disease, reporting protection of the endothelial glycocalyx and reduced albuminuria with OVZ/16-35. The authors note that pharmacokinetics, bioavailability, and long-term safety remain under study and disclose patents related to the compound, so this is promising animal evidence rather than clinical validation. These studies indicate therapeutic breadth, but differences in disease mechanism, intervention timing, administration route, and outcome choice preclude simple efficacy rankings.

Animal efficacy is most informative when dose, exposure, and a proximal HPSE pharmacodynamic measure are reported alongside the disease outcome. Without that linkage, an animal phenotype may support therapeutic potential but cannot resolve whether the compound reached the target tissue or acted through HPSE alone; comparisons should therefore remain specific to each model and modality.

4.2 Human pharmacology and clinical outcomes

Human studies provide the most consequential check on preclinical assumptions. Basche et al. [6] examined PI-88 in advanced solid tumors, establishing an early clinical pharmacology and tolerability foundation. Hudachek et al. [7] later developed a population pharmacokinetic model for PI-88, illustrating how systemic handling can be assessed quantitatively rather than inferred from molecular size or charge. A phase I roneparstat investigation in multiple myeloma reported tolerability within the studied dose levels, but early safety observations did not demonstrate established clinical efficacy.

The phase I pixatimod study enrolled 23 patients with advanced solid malignancies. The dose escalation identified 100 mg weekly as the maximum tolerated dose in the reported regimen; no objective responses were observed, although stable disease was described in some participants [9]. These observations establish real systemic exposure and a human safety boundary but should not be described as definitive evidence that HPSE inhibition lacks therapeutic value in every disease or treatment combination.

The randomized phase II necuparanib trial is particularly important because it tested whether the addition of a heparan sulfate mimetic improved outcomes when combined with gemcitabine and nab-paclitaxel in untreated metastatic pancreatic ductal adenocarcinoma. Median overall survival was 10.71 months with necuparanib and 9.99 months with placebo (hazard ratio, 1.12; 95% CI, 0.66–1.89; $P = 0.671$), with the point estimate favoring placebo. Median progression-free survival was 5.52 versus 6.93 months, respectively (hazard ratio, 0.97; 95% CI, 0.61–1.54; $P = 0.886$), and response rates were comparable. The report is internally inconsistent about enrollment: its abstract says 110 patients were randomized, whereas the Results section reports 120 (62 assigned necuparanib and 58 placebo), consistent with the arm counts. The trial crossed a planned futility boundary and did not demonstrate improved overall survival [10]. This is stronger negative clinical evidence than an unsuccessful preclinical optimization, although attribution of the result to inadequate target inhibition, disease biology, or intervention context requires additional data.

That clinical result should be interpreted alongside, but not displaced by, later preclinical work. Chen et al. [48] reported that PI-88 suppressed stromal activation in coculture and, with gemcitabine and nab-paclitaxel, improved tumor control and survival in orthotopic PDAC models. The study supplies a mechanistically developed combination rationale, but remains preclinical, concerns PI-88 rather than necuparanib, and reports the underlying data as confidential. It therefore does not resolve the negative phase II result or establish clinical benefit.

These outcomes do not support a simple contrast in which clinical mimetics failed and small molecules are therefore intrinsically superior. The trials establish that systemic administration and tolerability can be characterized, and necuparanib provides randomized negative efficacy evidence in its tested regimen; they do not, across programs, provide a comparable target-site HPSE pharmacodynamic measure that links exposure to enzyme suppression and outcome. Consequently, the necuparanib result does not identify whether inadequate target inhibition, disease context, combination strategy, or HPSE-independent pharmacology explains the outcome. Future trials should pre-specify a proximal, indication-relevant pharmacodynamic

marker and assess its relationship to exposure and clinical endpoints. The distinct human evidence for PI-88, roneparstat, pixatimod, and necuparanib is summarized in Table 2.

Table 2: Clinical studies of heparan sulfate mimetics.

Program	**Phase and setting**	**Reported evidence**	**Interpretation for lead development**
PI-88 / muparfostat	Phase I, advanced solid tumors [6]	Human biological/pharmacologic evaluation; subsequent population PK modeling [7]	Human exposure is measurable; efficacy requires separate controlled tests
Roneparstat / SST0001	Phase I, multiple myeloma [8]	Acceptable tolerability in tested dose cohorts	Early safety does not demonstrate efficacy
Pixatimod / PG545	Phase I, 23 advanced solid-tumor patients [9]	MTD 100 mg weekly in the studied regimen; no objective responses	A defined clinical exposure and tolerability benchmark with limited antitumor response
Necuparanib / M-402	Randomized phase II metastatic pancreatic cancer [10]	OS 10.71 vs 9.99 months (HR 1.12); PFS 5.52 vs 6.93 months (HR 0.97); comparable response rates. Abstract reports 110 randomized; Results report 120 (62 vs 58)	No OS or PFS benefit; enrollment total is internally inconsistent in the report

5 Computational discovery and emerging modalities

5.1 Computational discovery: hypothesis generation versus validation

Ligand-based screening, structure-based docking, molecular dynamics, free-energy calculations, and QSAR models are prominent across HPSE discovery studies. Gozalbes et al. [13] combined structure- and ligand-based hit identification with experimental assays. Pala et al. [16] used iterative similarity screening followed by chemical testing. Such workflows are most persuasive where prediction produces independently tested molecules and where the authors report inactive as well as active compounds.

Shanbhogue et al. [46] provide a recent example of machine-learning-based HPSE activity classification. Their model used 225 curated ChEMBL measurements and a stratified random 80:20 training/test split, reporting 80% overall accuracy and 78.5% balanced accuracy on the holdout set. The authors also predicted activity classes for 14 compounds outside the training/test data and ranked 900 database-screened candidates, but did not report biochemical testing of these proposed hits. They further state that the large-library screen used 2D descriptors and Morgan fingerprints alone, although the selected model was trained with an additional 3D descriptor set; clarification of how this feature-space difference was handled would improve reproducibility. The study therefore advances computational prioritization, not experimentally confirmed discovery of new HPSE inhibitors.

Zhang et al. [26] offers a central cautionary case. The biochemical hit TC LPA5 4 inhibited HPSE with IC_{50}

= 10 μM. A designed analog, MX4-62, was synthesized but showed no measurable inhibitory effect in the authors' assay. The study then used docking, a 500 ns simulation for the hit, a corresponding simulation for MX4-62, and simulations of further designed analogs to propose structural explanations. The inactivity of MX4-62 was an experimental finding within the same paper, whereas much of the expanded SAR proposal remained computational. These two evidence levels should not be merged into an assertion of validated design rules.

Docking scores are not direct measures of IC_{50}, and persistence of contacts in a simulated complex does not establish binding affinity, selectivity, or functional inhibition. These methods are useful when they generate a falsifiable medicinal chemistry choice: define the predicted interaction or rank order, synthesize a small matched set including a negative control, and test it prospectively with an orthogonal enzyme assay and, where warranted, a binding or cellular assay. Retrospective machine-learning performance should be interpreted in light of scaffold separation, assay heterogeneity, activity cliffs, applicability domain, and feature consistency; prospective confirmation is the step that tests whether a model informs new chemistry. The inactivity of MX4-62 [26] illustrates the value of reporting negative as well as positive outcomes.

5.2 Emerging HPSE-targeting modalities and boundaries of this review

Conventional catalytic inhibition is not the only way to modulate extracellular HPSE biology. Li et al. [44] reported a heparanase-directed lysosome-targeting chimera (LYTAC), designated JW-9, intended to promote extracellular HPSE degradation and improve natural killer cell antitumor effects in experimental models. This strategy changes the therapeutic question from transient inhibition of a catalytic site to reduction of extracellular protein abundance. It is appropriately discussed as an emerging comparator rather than counted as a conventional small-molecule active-site inhibitor.

Other agents modify heparan sulfate availability or growth-factor recognition without necessarily demonstrating selective HPSE catalysis blockade. Such studies may inform off-target liabilities or the broader extracellular matrix context, but they should not be included in a primary biochemical-inhibitor table unless direct HPSE activity has been demonstrated. This boundary is essential for preventing overstatement of the chemical evidence base.

6 Lead appraisal and future directions

6.1 Evidence-based appraisal of lead quality

The literature supports a useful distinction between a biochemical hit, a mechanistically supported inhibitor, a biologically validated lead, and a pharmacologically characterized candidate. These labels should be assigned using study-specific evidence rather than arbitrary IC_{50} cutoffs. An IC_{50} below 100 nM may be highly desirable, yet an unselective or poorly characterized compound can remain a hit. A micromolar compound may be a productive lead if its mechanism, selectivity, formulation, exposure, and cellular response are well supported.

A useful appraisal combines a reproducible, substrate-defined enzyme result with orthogonal confirmation

and selectivity tests appropriate to the chemotype. Mechanistic binding evidence, cellular HPSE dependence, exposure-linked pharmacodynamics, safety, and disease-relevant outcomes add confidence when they address the claim being made; they are complementary domains, not a mandatory experiment sequence. Highly sulfated glycans, covalent inhibitors, degraders, and conventional small molecules require different selectivity and engagement measurements.

Figure 1 maps these evidence domains around claim-specific appraisal rather than arranging them as a fixed progression. Figure 2 provides a worked example of how matched structural comparisons can reveal context-dependent potency changes. Neither figure assigns a validated numerical score.

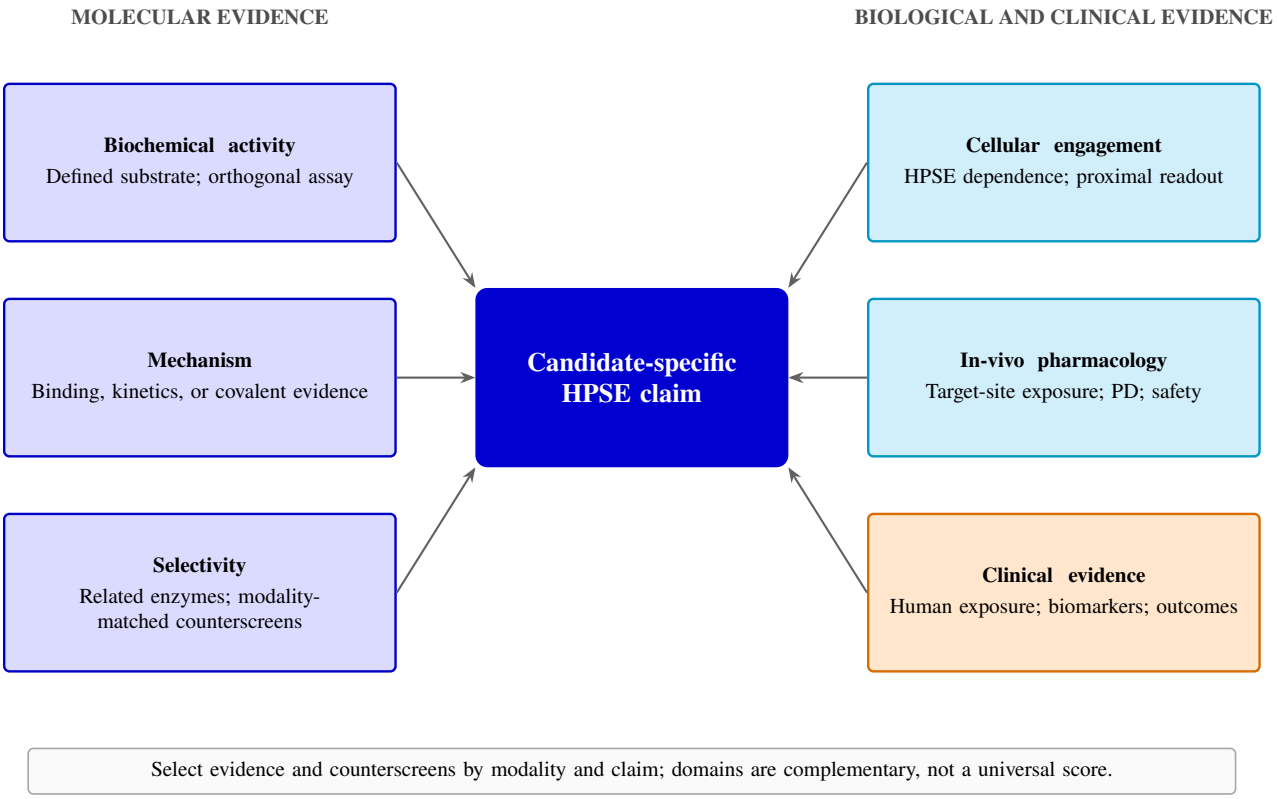


Figure 1: Non-linear evidence map for evaluating HPSE-directed candidates. Molecular, cellular, in-vivo, and clinical evidence inform candidate-specific claims from complementary directions; the relevant experiments and counterscreens depend on modality. The map is conceptual, not a validated score or mandatory development sequence.

The framework supports comparison without imposing a maturity ladder. Messore's 7g has strong within-assay potency and selected cellular phenotypes, but limited direct cellular target-engagement evidence; the Imai series combines selectivity optimization with serum HPSE pharmacodynamics, but not demonstrated diseased-tissue engagement or efficacy. Davies' fragments have experimentally observed binding structures and micromolar enzyme inhibition but remain early in cellular and in-vivo characterization. De Boer's cyclophellitols combine mechanism-based inhibition with mouse outcomes, whereas clinical glycomimetics offer human exposure and tolerability data without consistent efficacy. These profiles answer different questions and cannot be ordered by a universal score. Table 3 distinguishes what each evidence type establishes from what it cannot establish alone; Figure 2 shows how matched structural changes can also reverse apparent SAR within one chemotype.

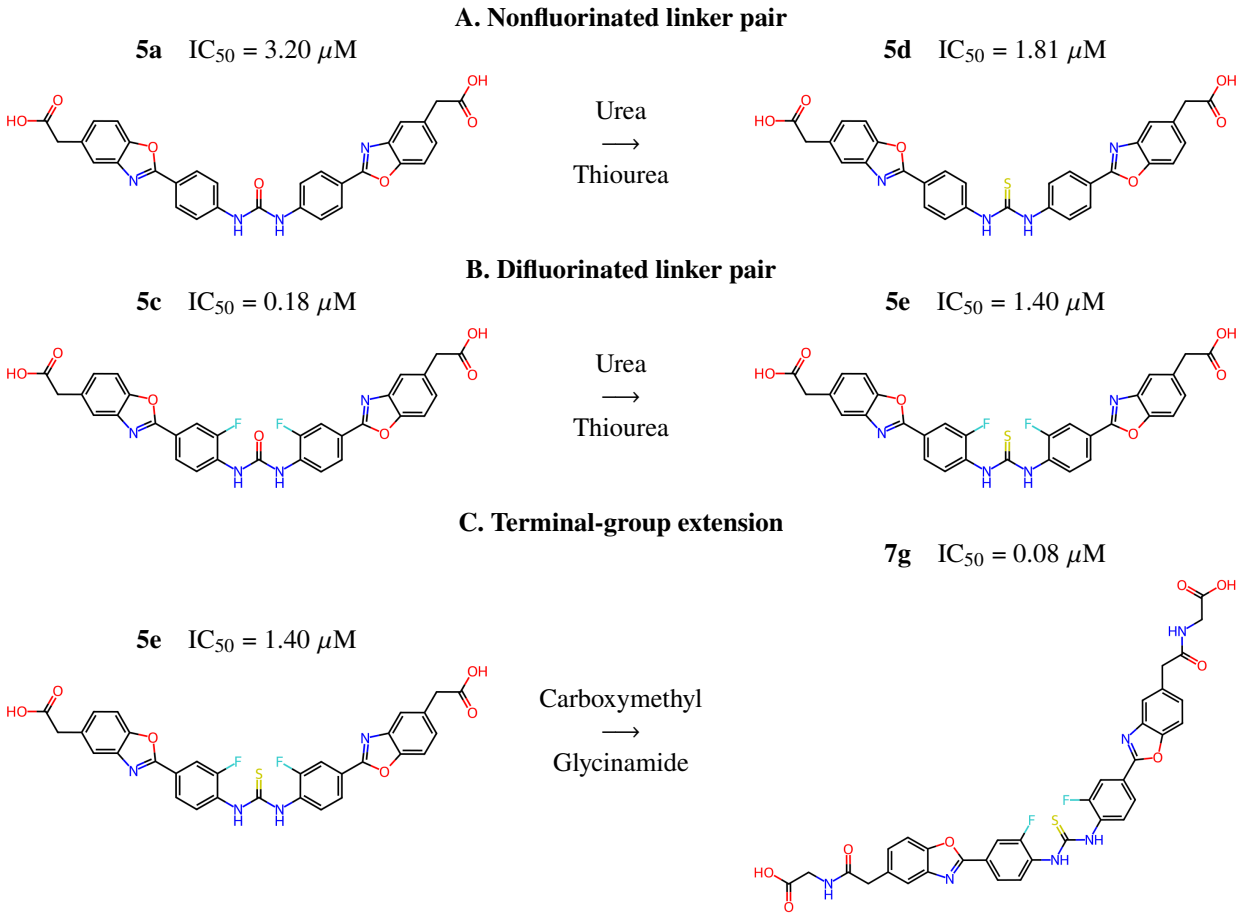


Figure 2: Structure–activity relationships for selected symmetrical benzazolyl HPSE inhibitors reported by Messore et al. [15]. In A, replacing urea with thiourea is associated with a 1.8-fold potency gain in the nonfluorinated pair; in B, the same linker change is associated with a 7.8-fold loss in the difluorinated pair. C compares 5e with the glycinamide-extended analogue 7g, whose IC_{50} is 17.5-fold lower in the same fondaparinux-based assay. Structures and assay values were checked against the source-reported compounds and assay in Messore et al.; values are not cross-study potency rankings.

Table 3: Evidence domains for evaluating heparanase inhibitors.

Evidence domain	Typical measurements	What it establishes	What it does not establish alone
Biochemical inhibition	IC_{50}, K_i, inhibition kinetics under stated substrate conditions	Effect on a defined enzyme assay	Whole-organism potency or mechanism specificity
Assay confirmation	Orthogonal enzyme assay, interference control, repeatability	Greater confidence that the biochemical effect is genuine	Cellular target engagement
Selectivity	GUSβ/GBA counterscreens; PF4, antithrombin, growth-factor binding panels	Reduced activity at chosen alternative targets	Complete proteome-wide selectivity or safety
Structural mechanism	Cocrystal structure, kinetic model, covalent labeling, biophysical binding	More direct explanation of enzyme–ligand interactions	Effective dose or disease benefit
Cell-based relevance	HPSE activity in cells, invasion, matrix degradation, genetic/rescue controls	Relevant cellular pharmacology, strongest with mechanistic controls	Systemic exposure or clinical efficacy
Animal validation	PK/PD relationship, tumor growth, metastasis, inflammation, survival	In-vivo proof of concept in a specified model	Efficacy in humans
Clinical development	Exposure, tolerability, response rates, randomized outcomes	Human safety and/or efficacy under tested regimen	Transferability to other indications

6.2 Research gaps and future directions

Assay comparability: HPSE inhibitor programs would benefit from a common reference panel tested in parallel using at least two orthogonal substrates. Reference compounds should include distinct chemotypes to determine whether assay behavior is broadly transferable or scaffold dependent. Shared reporting of enzyme construct, active enzyme fraction, substrate identity, kinetic regime, readout interference checks, and curve-fitting uncertainty would make published IC_{50} values more interpretable.

Mechanistic specificity: Catalytic and noncatalytic HPSE effects should be distinguished in cellular systems. Whole-cell phenotypes should be interpreted alongside direct enzyme activity, target engagement, and controls that separate HPSE inhibition from growth-factor sequestration or general extracellular matrix disruption. For glycomimetics, systematic binding panels are especially relevant; for covalent inhibitors, proteome-level reactivity and enzyme-family selectivity require attention.

Medicinal chemistry decisions: Prospective matched-pair analog series should be prioritized over retrospective interpretation of a few potent compounds. Chemical synthesis practicality, solubility, protein binding, formulation, and stability should be measured before extensive animal studies. Computational SAR should be labeled as predictive until a newly designed compound has been experimentally tested. Negative analog results, exemplified by MX4-62, are highly informative and should be published rather than omitted.

Exposure-response alignment: Demonstration of target-site concentration and pharmacodynamic HPSE inhibition should accompany mouse efficacy whenever feasible. The clinical mimetic experience suggests

that acceptable tolerability and systemic exposure alone do not guarantee a useful therapeutic window. Explicit disease-context selection, biomarker strategy, combination rationale, and clinically relevant dosing are needed to translate broad target biology into tractable therapeutic hypotheses.

Therapeutic positioning: The most productive indications may be those in which extracellular HPSE dependence is measurable, intervention timing is clear, and a proximal pharmacodynamic biomarker is accessible. Oncology remains important, but sepsis, glycocalyx dysfunction, and fibrosis provide mechanistically distinct opportunities. Evidence should not be combined across those indications without accounting for route of administration, disease stage, and the role of concurrent inflammation or signaling pathways. Practical study-design and reporting priorities are summarized in Table 4.

Table 4: Concise reporting checklist for heparanase inhibitor studies. This table lists reproducibility fields; Table 3 separately explains what each evidence domain can establish.

Reporting package	**Study-specific information to report**
Enzyme assay	Enzyme construct and maturation state; substrate identity and concentration relative to kinetic conditions; incubation and readout; replicate number, raw dose-response data, and fitting uncertainty
Assay confirmation	Orthogonal detection chemistry; optical, aggregation, and reporter-interference controls; matched positive and negative compounds
Selectivity panel	Comparator enzymes or heparan sulfate-binding proteins selected for the chemotype, with concentrations and assay conditions
Mechanism	Kinetic model and evidence for reversible, covalent, allosteric, or abundance-modulating action; distinguish measured binding from predicted poses
Cellular studies	Cell exposure and viability; proximal HPSE activity or abundance readout; genetic, rescue, or structurally distinct inhibitor controls where available
In-vivo studies	Dose, route, formulation, time-matched plasma and target-tissue exposure, pharmacodynamic readout, safety measures, and efficacy endpoints
Clinical studies	Dose and exposure; prespecified target-proximal biomarker and sampling time; eligibility criteria, treatment context, and clinical endpoints

7 Limitations of the evidence synthesis

As a narrative review based on focused searches, this synthesis may not include every relevant study. Representative studies were selected to illustrate distinct chemical and translational evidence profiles, not to estimate the relative frequency or overall success of each inhibitor class. Variation in assay methods and reporting limits comparisons of potency, target engagement, and animal outcomes; some mechanistic interpretations also depend on experimental details unavailable in the examined sources. The conclusions therefore represent a critical interpretation of selected literature, not an exhaustive ranking of HPSE inhibitor classes.

8 Conclusions

The most informative medicinal chemistry result in the reviewed benzazolyl series is not a single low IC_{50} but the reversal of the urea-to-thiourea effect between fluorinated and nonfluorinated analogues; the glycine-amide extension in 7g adds a second, experimentally measured within-series gain. More broadly, lead evidence remains complementary rather than linear: Imai compounds combine selectivity optimization with serum pharmacodynamics, cyclophellitols combine mechanism-based inhibition with animal outcomes, Davies' fragments provide structural binding evidence but little downstream pharmacology, and clinical glycomimetics establish human exposure and tolerability without consistent efficacy. These records do not support a universal chemotype ranking. The next decisive advances are prospective, assay-matched SAR; modality-appropriate selectivity and causal cellular tests; and exposure-linked, indication-specific pharmacodynamics. A candidate should advance when its specific mechanistic and therapeutic claims are supported by convergent evidence, not because it clears a potency threshold.

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